

# Memo Ozdincer

**Built** a weight-level LLM-agent injection defense (NeurIPS, COLM submitted, first author). **Sped-up** solar-cell processing 100,000x at NUS's SERIS lab. **Developed** molecular-dynamics chemical diffusion model at Matter Lab × NVIDIA.

University of Toronto, BAsC in Engineering Science (Eng. Physics + AI minor) - Dean's list (3.7 GPA), Jun 2028

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## Experience

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**Jinesis AI Lab (Vector Institute)** - AI Researcher (Safety)

Aug 2025 – Present

**Advisors:** Zhijing Jin, Bernhard Schölkopf

- Designed a weight-level LLM agent defense competitive with Meta SecAlign, OpenAI IH, Google CaMeL, and MELON on the ASR-utility Pareto across standard agentic benchmarks (<0.8% ASR on AgentDojo, InjecAgent). First-author NeurIPS submission.
- Built a 100K-trace dataset of entirely benign-looking prompt injections, plus a 566K JEPA-format training set, reverse-generating prompts from 15 jailbreak datasets × 20 augmentation methods (COLM, NeurIPS).
- Rolled out and evaluated >1M traces (1B+ tokens) across Llama, Qwen, Mistral, DeepSeek, and Gemma models (8B–685B) in Triton and TensorRT-LLM. Parallelized across 4xH100, 8xH200, and 2x/4xB200 nodes on bare metal/NVLink, never letting GPU utilization drop below 90%.

**Matter Lab (Vector Institute + NVIDIA)** - AI Researcher (Generative Chemistry)

Sep 2025 – Present

**Advisor:** Alán Aspuru-Guzik

- Beat Sella, the status-quo saddle-search algorithm, on highly noisy atomic potentials: 79% convergence at 1Å noise vs. Sella's 53%. Submitting to NeurIPS 2026.
- Redesigning search algorithms to be differentiable for generative model training, enabling adjoint backpropagation to train diffusion models on our custom vector field.
- Parallelized over 3.8M molecular dynamics simulations across 12 nodes of 4× NVIDIA A100 SXM4 (NVLink) using our HIP MLIP model, developed with NVIDIA.

**National University of Singapore (SERIS)** - ML Researcher (Comp. Physics)

May 2025 – Aug 2025

**Advisors:** Erik Birgersson, Armin Aberle

- Designed Physics-Informed ML model for solar cell characterization, replacing a slow coupled PDE solver (~4800 s/device) with millisecond inference, a 100,000x speedup. Deployed at SERIS for high-throughput screening.
- Trained on 500K simulations (31 quantum solar-cell properties spanning 21 OOM). Hand-derived physics priors from 71 quantum and optoelectronic relations.

**aUToronto (UofT AutoDrive Team)** - Software Engineer (Computer Vision)

May 2025 – Present

- Implemented BEVFusion-based sensor fusion pipeline (C++/Python) processing 1.1 GB/s from 6 cameras + 2 LiDARs at under 90ms latency for real-time tracking.
- Fine-tuned lightweight detection models (Twin+, YOLOPV2, YOLO11) with 35% perception speedup in adverse weather. Wrote LiDAR–camera calibration tooling in C++, reducing false-positive edge detections by 85%+.

## Skills

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**AI/ML:** LLM & Agent Safety, Fine-Tuning, Evals, Reinforcement Learning, Diffusion Models, CV (YOLO, BEVFusion)

**Languages & Frameworks:** Python, C++17, C, CUDA, PyTorch, vLLM, Triton, TensorRT-LLM, Hydra/OmegaConf, DVC, W&B, Tensorflow, Docker, Kubernetes, HuggingFace, AWS, SLURM/HPC